Naïve Bayes Theorem

Bayesian Concept Learning

- In Bayesian learning, we consider: A **hypothesis space** H
- Each hypothesis has a probability
- We compute:
  - $P(h \mid D) = \frac{P(D|h)\ P(h)}{P(D)}$
- Where:
  - h: Hypothesis
  - D: Training data
  - $P(h \mid D)$:Posterior probability of hypothesis

Bayesian Concept Learning

- **Selecting Hypothesis**
- **MAP Hypothesis (Maximum A Posteriori):**
- $h_{MAP} = \arg\max_{h} \quad P(h \mid D)$
- **ML Hypothesis (Maximum Likelihood):**
- $h_{ML} = \arg\max_{h} \quad P(D \mid h)$

Concept	ML	MAP
Uses data?	Yes	Yes
Uses prior knowledge?	No	Yes
Thinking style	“What looks most likely?”	“What makes most sense overall?”

Applications of Bayes' Theorem

- **1. Spam Email Filtering:** Classifies emails as **Spam / Not Spam**
- Uses word probabilities
- Example: "Free", "Offer" → high spam probability
- **2. Medical Diagnosis:** Calculates probability of disease given symptoms
- Used in clinical decision systems
- **3. Fraud Detection:** Detects unusual transactions
- Updates probability with each new activity
- **4. Recommendation Systems:** Suggests movies/products based on user behavior
- **5. Weather Prediction:** Updates forecasts based on new atmospheric data
- **6. Machine Learning (Naive Bayes Classifier)**
- Text classification , Sentiment analysis , Document categorization

1. Key Terminologies

Experiment: A process that produces outcomes
Example: Tossing a coin

- **Sample Space (S)**: Set of all possible outcomes
Coin: S={H,T}

- **Event (E)**: Subset of sample space
Getting Head: E={H}

2. Basic Probability Formula

$$P(E) = \frac{Number\ of\ favorable\ outcomes}{Total\ outcomes}$$

Example:
Probability of getting Head = $1/2$

3. Important Rules

(a) Range Rule

$$0 \leq P(E) \leq 1$$

(b) Complement Rule

$$P\left(\bar{E}\right) = 1 - P(E)$$

(c) Addition Rule

- For **mutually exclusive events**:

$$P(A \cup B) = P(A) + P(B)$$

- For **non-mutually exclusive events**:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

1. Key Terminologies

4. Conditional Probability

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Meaning: Probability of A **given that B has occurred**

5. Multiplication Rule

$$P(A \cap B) = P(A \mid B)\, P(B)$$

.6 Independent Events

Two events are independent if:

$$P(A \cap B) = P(A)\, P(B)$$

Formula

$$P(H \mid E) = \frac{P(E \mid H)\, P(H)}{P(E)}$$

Prior Probability is Known or Estimated

- Requires $P(H)$)prior)
- Often estimated from training data

Likelihood is Computable

- Must be able to calculate $P(E \mid H)$
- Depends on data type (categorical, continuous)

Evidence is Non-Zero

$$P(E) \neq 0$$

Otherwise, division becomes undefined.

1. Key Terminologies

Feature Distribution Assumption (in variants)
- Gaussian Naive Bayes → assumes normal distribution
- Multinomial → assumes discrete counts
- Bernoulli → assumes binary features